

PROPOSAL TO THE CPO & GMS OF PRODUCT

Measuring What Matters

A value-realization framework for product adoption in a
consumption-based security business

Principal PM, Product Adoption Analytics

Built end-to-end with a spec-driven, AI-first development workflow

CVRS

The Technical PM operating model

Specs are the contract

Markdown specs define the metric, data model, and pipeline first; code and tests are generated from them and kept in sync

Ship a prototype, not slides

Everything here runs: real BigQuery, a layered SQL pipeline, 28 automated tests, and a live Streamlit dashboard

Measure like an engineer

Every claim is reproducible and test-backed; the metric is evaluated the way a model would be, against labeled ground truth

Own the whole seam

Bridge Product, CS, and Finance - define the number, defend the trade-offs, and stand behind the implementation

CONTEXT & PROBLEM

Raw usage is lying to us

The task: build a framework to measure product adoption and customer value - scoring it at the customer, SKU, and feature level and classifying every account into an actionable health tier.

10%

pure shelfware - paying, zero consumption, invisible until renewal

5%

burn ~90% of entitlements in 3 months, then go dark - today scored as heroes

15%

run at 120%+ every month - expansion revenue nobody systematically works

Why leadership cares

- Renewal surprises: churn risk hides behind cumulative usage charts
- Wrong incentives: teams paid on volume they can game
- SKUs bundle many features - we measure the bundle, not whether value inside it is realized

One score, four honest components

40%

Consumption utilization

Consumed vs entitled, capped at 100% - burning faster is not healthier

30%

Feature adoption depth

Features used vs eligible, per SKU - catches shallow deployments

20%

Consistency

Trailing 3-month trend vs peak - punishes spike-and-drop within a month

10%

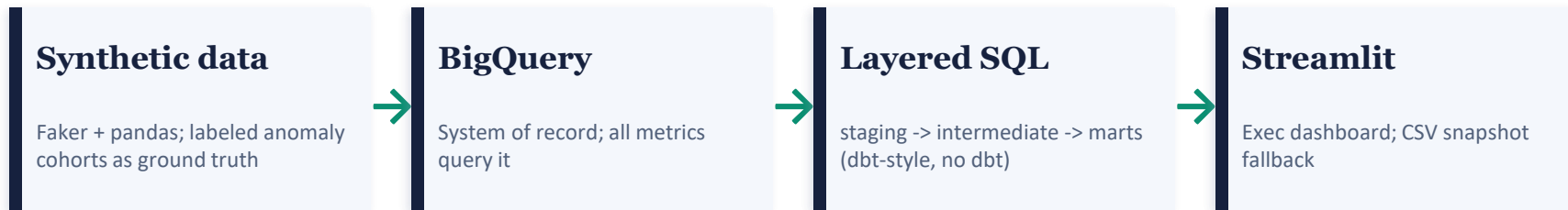
Breadth

SKUs with real usage vs owned - no credit for ignored products

Health tiers drive playbooks: **Healthy 70+** **Watch 40-69** **At Risk <40 or drop** **Shelfware = zero use**

Revenue overlay: committed ARR, ARR-at-risk, and expansion pipeline sit alongside the score - never inside it (utilization is capped; a test proves CVRS depends only on its four components).

From synthetic data to executive dashboard



Build vs Buy

Buy the warehouse (BigQuery); build the transforms as plain layered SQL - dbt's discipline without its dependency for a prototype

Latency vs Accuracy

Monthly batch is the score's grain - entitlements are monthly and QBR/renewal decisions are monthly

Two speeds, one metric

A daily consumption feed powers a pace-adjusted, projected CVRS for the in-flight month - leaders see it unfolding, without changing the monthly grain

The hard part solved in SQL

Mid-year expansions summed per month via a date-range join - overlapping contracts raise the denominator, no double-counting

Designed for messy enterprise data

Spike & drop

Consistency compares this month to the trailing 3-month peak - a cliff to zero collapses the score within a month

Shelfware

Reporting is built from entitlements, not usage - zero-consumption accounts can never silently vanish

Consistent overage

Utilization capped in the score; raw overage feeds the expansion pipeline instead of inflating health

Mid-year expansion

Denominator steps up in the expansion month; a 2-month grace window stops the expected dip reading as churn

Deployment lag

A real per-product adoption date drives day-level time-to-first-value (0-90 days), not a monthly approximation

We evaluated the metric like a model

Ground-truth validation

- Synthetic data ships with labeled cohorts - the answer key is known before scoring
- Tests require $\geq 90\%$ of each injected cohort to land in the intended health tier
- 28 automated tests: structural integrity, score bounds, no-account-dropped, and cohort detection - green on every run
- BigQuery caught real bugs a local run hid: a CTE/column name collision and negative time-to-value

Tied to business KPIs

Shelfware tier

Deployment rescue -> time-to-first-value

At Risk tier

Save motions -> gross retention

Expansion flag

True-up pipeline -> net revenue retention

Feature depth

Activation campaigns -> roadmap bets

Hard truths we kept the receipts for

Pipeline & data

CTE / column name collision

BigQuery averaged a whole row-struct instead of the score - it passed my local logic and surfaced only against the real warehouse's stricter engine

Negative TTFV for 82% of accounts

Measuring day-precision against monthly consumption; fixed with a real product-adoption date, not arithmetic on mismatched grains

A dormant safeguard

The relative-drop At-Risk rule never independently fires - the absolute floor catches every decline first. Documented as a known gap, not hidden

Dashboard & delivery

Clickable executive KPIs

Streamlit metrics can't be links, so tier percentages became theme-aware HTML cards that jump to the named customers behind them

A chart execs couldn't read

The time-to-adopt box plot (quartiles, whiskers) became one labeled bar per platform - the message reads without a tooltip

Wrong month labels

Plotly auto-placed datetime ticks; pinned to explicit month-name labels so the trend axis is exact and ordered

The road ahead

If rebuilt with today's stack

- Agentic triage: an agent investigates every tier change and drafts the CSM brief
- LLM-generated account narratives on top of the component data - the 'why', not just the score
- Keep the spec-driven loop: metric changes ship as spec diffs, with code and tests regenerated from them

What I'd fund first

- The shadow quarter: run CVRS silently on real telemetry and back-test - does the month-9 score predict the month-12 renewal?
- Only then productionize: warehouse-native pipeline and BI integration
- Charter the metric governance forum (Product + CS + Finance)

The ask: endorse CVRS + health tiers, fund the shadow quarter, charter governance. *Live demo next.*

How CVRS is calculated

$$\text{CVRS} = 100 \times (0.40 \cdot \text{utilization} + 0.30 \cdot \text{feature depth} + 0.20 \cdot \text{consistency} + 0.10 \cdot \text{breadth})$$

Worked example — **Brooks, Lam and Hayes** (Mid-Market · AMER · Jun 2026)

Component	What it measures	Value	Weight	Points
Consumption utilization	consumed ÷ entitled, capped at 100%	9,179 / 12,906 = 71%	40%	28.4
Feature adoption depth	features used ÷ eligible, per SKU	75%	30%	22.5
Consistency	retention vs trailing-3-month peak	0.87	20%	17.4
Breadth	SKUs with usage ÷ SKUs owned	100%	10%	10.0
CVRS (sum of points)				78.3

→ 78.3 lands in the **Healthy** tier (≥70). Tiers: Healthy ≥70 · Watch 40–69 · At Risk <40 or a ≥25-pt drop · Shelfware = zero use.